

# Dailin Song

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## EDUCATION

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|------------------------------------------------------------------------------|-------------------------------------|
| <b>Columbia University</b><br><i>Master of Science in Statistics</i>         | New York, NY<br>09/2024 - 12/2025   |
| <b>University of Wisconsin</b><br><i>Bachelor of Science in Data Science</i> | 09/2022 - 05/2024<br>GPA: 3.61/4.00 |

## SKILLS

**Data Analytics & Visualization:** SQL, Python (Pandas, NumPy, Matplotlib), R, Power BI, Tableau, Excel (vlookup, pivot table)

**Database:** MySQL, PostgreSQL, SQL Server, MongoDB, Oracle

**Certification:** Society Of Actuaries Probability (P) Exam

**Statistics:** Linear Regression, Hypothesis Tests, Probability Theory, Bayesian Inference, Maximum Likelihood Estimation

**Machine Learning Techniques:** TensorFlow, Decision Tree, Random Forest, XGBoost, LSTM, K-Nearest Neighbors, CatBoost

## PROFESSIONAL EXPERIENCE

**WeSpark LLC** | *Data Analyst* | Atlanta, GA 06/2025 – 08/2025

- Optimized workflows by restructuring and merging data, reduced data analysis processing time by 10%, and cutting costs by 5%
- Developed Python scripts to analyze procurement data and detect inconsistencies in output reports, accelerated data **ETL** process
- Integrated industry benchmarks in **API calls**; cleaned data to compute category averages and parameterized model assumptions
- Designed an end-to-end indirect procurement workflow in Visio, mapping Category Management and Tactical Buy processes from supplier onboarding to payment, reducing buyer-cycle time by 53%

**PFPA Financial Consulting** | *Data Science Intern* | Atlanta, GA 12/2024 – 02/2025

- Extracted and aggregated 2-year daily stock prices and quarterly financial statements for 20+ public companies with **Bloomberg**
- Modeled firm credit risk** using Merton's structural framework by treating equity as a call option on assets and applied Expectation-Maximization (EM) algorithm to infer latent variables such as asset volatility and value
- Assessed default risk** by calculating Distance-to-Default (DD) and Expected Default Frequency (EDF), and converted results into Time-Consistent (TiC) scores for through-the-cycle (TTC) credit ratings
- Conducted **cross-industry risk analysis**, identifying stable credit patterns in manufacture industry (Coca-Cola) vs. techs (Intel)
- Visualized quantitative credit ratings and risk patterns for data-driven investment decisions, improved client advisory outcomes

**Yinhua Fund Management Co., Ltd.** | *Data Science Intern* 05/2024 - 08/2024

- Analyzed CSI 300 Index and volume data, and conducted **EDA** and applied indicators (EMA, SMA, RSI) to detect trading patterns
- Implemented **Fama-French Three-Factor Model** using **Python** to estimate stock excess returns on market-cap and value factors, quantifying exposure to risk factors
- Incorporated results with fundamental metrics, and applied Z-score standardization to generate a composite growth score
- Ranked Top 100 growth stocks by average Z-score and delivered results to Quant Team to support portfolio construction strategies

**iFLYTEK Co., Ltd.** | *Operations Analyst Intern* 05/2023 - 08/2023

- Conducted **operational data analysis** to identify bottlenecks in banking service workflows, reducing call center labor by 27%
- Designed and implemented a conversational AI chatbot trained on internal banking SOPs to **automate customer service tasks**
- Improved customer satisfaction scores by 15% and **reduced service time** by 1 hour via data-informed automation and integration
- Collaborated with cross-functional teams to ensure AI solutions met real-time needs and scaled across service channels

**Credera** | *Data Analytics Intern* 05/2022 - 07/2022

- Collected and analyzed 3,000+ responses to GenZ talent preferences regarding employers, incentives and workplace expectations
- Designed and maintained a **SQL** database to manage incoming survey data and ensure efficient querying, updating, and reporting
- Performed ongoing data maintenance by updating, adding, and removing entries as new responses for data quality and accuracy
- Produced key insights to HR team to inform recruitment strategies and align talent acquisition efforts with Gen Z expectations

## PROJECTS

**Credit Card Fraud Test** 05/2025 - 07/2025

- Analyzed a dataset of 500,000+ transactions by **Data ETL** using **Python** and **SMOTE** to identify frauds, addressed class imbalance
- Trained and optimized various machine learning models, such as **Logistic Regression, Decision Tree, Random Forest, XGBoost, CatBoost, K-Nearest Neighbors**, with **GridSearchCV**, achieving a 97.6% accuracy with **XGBoost**
- Evaluated model performance through **Confusion Matrices** and **ROC-AUC** scores, ensuring robustness in fraud detection
- Simulated bank transactions to estimate model's impact, projecting a daily reduction of approximately \$200 in fraud-related losses

**Price Prediction and Risk-Adjusted Portfolio Optimization in Cryptocurrencies** 01/2025

- Analyzed the integration of LSTM networks and XGBoost for cryptocurrency price forecasting
- Authored a chapter published by IGI Global in *Machine Learning and Modeling Techniques in Financial Data Science* (DOI: 10.4018/979-8-3693-8186-1.ch013)